K.R. MANGALAM UNIVERSITY

Database Management Lab File (ENCS205)

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

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B. Tech CSE (Core) – 4th Semester

Section: A

Submitted To:

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EXPERIMENT: 2

2. **Consider following databases and draw ER diagram and convert entities and relationships to relation table for a given scenario:**

AIM: To analyze the requirements for a Company Database, design an Entity-Relationship (ER) Diagram representing Employees, Departments, Projects, and their relationships (Works On, Manages, Supervision), and map the ER diagram into a normalized Relational Database Schema.

PROCEDURE:

Step 1: Draw the 3 Main Entities (Rectangles)

Draw three rectangular boxes in the following positions:

1. DEPARTMENT: Place it in the top-middle.
2. Employee: Place it on the right side.
3. PROJECT: Place it in the bottom-left.

Step 2: Add Attributes (Ovals)

Draw ovals connected to their respective rectangles. Crucially, include the Foreign Keys as shown in your image:

- For DEPARTMENT:
  - Add ovals: DNAME, MGRSSN, MGRSTARTDATE.
  - Add oval DNO and underline it (Primary Key).
- For Employee:
  - Add ovals: NAME, ADDRESS, SALARY, SEX. o Add oval SSN and underline it (Primary Key).
  - Add Foreign Keys: Add ovals for SUPERSSN and DNO attached to this entity.
- For PROJECT:
  - Add ovals: PNAME, PLOCATION.
  - Add oval PNO and underline it (Primary Key).
  - Add Foreign Key: Add an oval for DNO attached to this entity.
- For Locations:
  - Draw a separate oval labeled LOOC below the Department table.

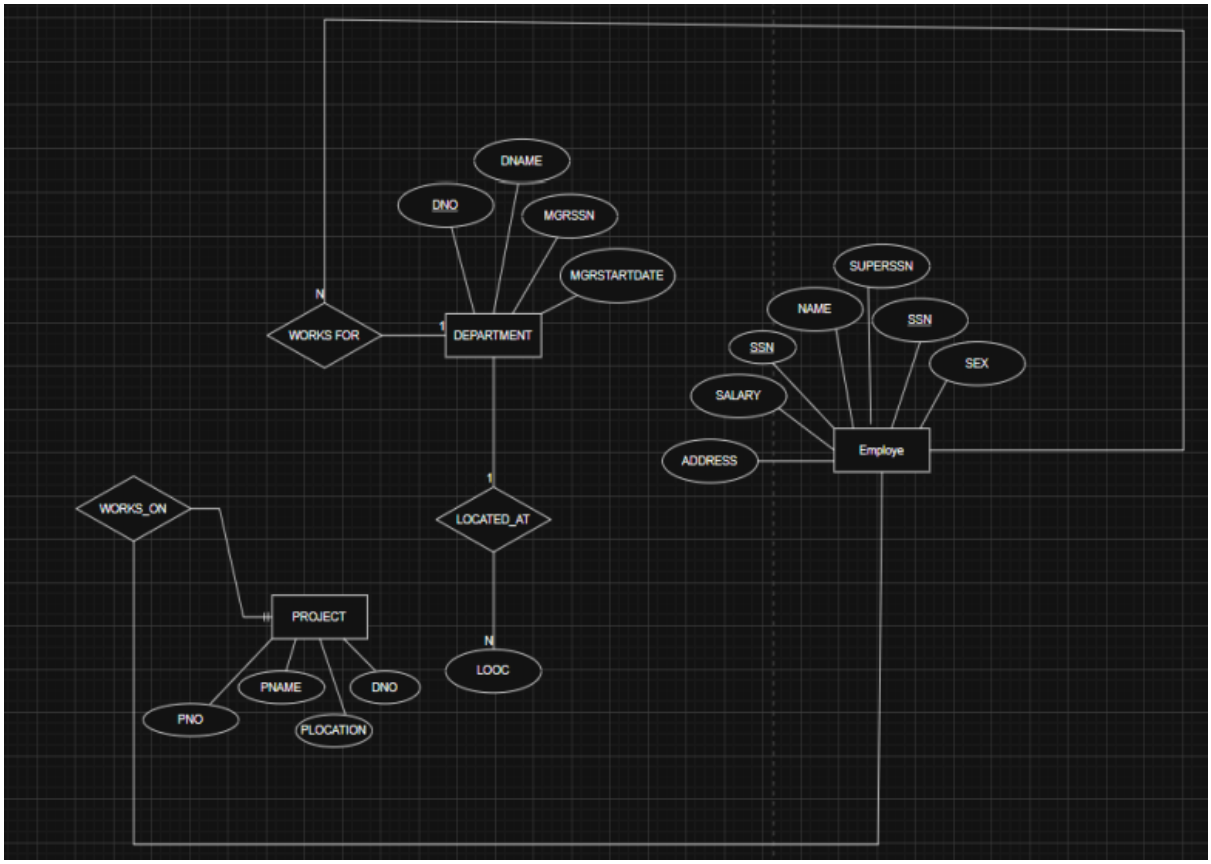
Step 3: Draw Relationships (Diamonds)

Draw three diamond shapes to represent the connections:

1. WORKS_FOR: Place this between DEPARTMENT and Employee.

2. LOCATED_AT: Place this between DEPARTMENT and the LOOC oval.
3. WORKS_ON: Place this to the left of the PROJECT table.

Step 4: Connect Lines & Define Cardinality

1. Connect WORKS_FOR:
   - Draw a line from DEPARTMENT to the diamond. Label it 1 (or N depending on your specific image logic, usually Many Employees work for 1 Dept). *In your image, N is near Employee and 1 is near Dept.*
   - Draw a line from Employee to the diamond**.**
2. Connect LOCATED_AT:
   - Draw a line from DEPARTMENT down to the diamond. Label it 1. ○ Draw a line from the diamond down to the LOOC oval. Label it N.
3. Connect WORKS_ON:
   - Draw a line from PROJECT to the diamond.
   - Draw a long line from the other side of the diamond that goes Up, turns Right (across the top of the page), turns Down, and connects to the Employee table.

This represents the Many-to-Many relationship between Employees and Projects